

LAB ASSIGNMENT – 6

Bias–Variance Tradeoff and Model Comparison

Q1. Bias–Variance Analysis in Polynomial Regression

Objective: To experimentally demonstrate the bias–variance tradeoff using polynomial regression models.

Dataset Generation

Generate a synthetic dataset with at least 700 samples:

$$x \sim U(-2, 2)$$

Target function:

$$y = \sin(\pi x) + 0.3x^2 + \epsilon$$

where

$$\epsilon \sim \mathcal{N}(0, 0.15)$$

Common Experimental Setting

- Train–test split: 70%–30%
- Optimization method: Gradient Descent
- Number of training epochs: 100 (fixed)
- Learning rate: Choose appropriately
- Metrics must be recorded at every epoch

Part A: Effect of Model Complexity

Construct polynomial regression models with:

$$d \in \{1, 3, 5, 7, 9, 11\}$$

For each degree:

1. Train for 100 epochs.
2. Record at every epoch:
 - Training MSE
 - Testing MSE

Epoch-wise Plot

For each polynomial degree d :

- x-axis: Training epoch
- y-axis: Mean Squared Error (MSE)
- Include two curves:
 - Training MSE vs epoch
 - Testing MSE vs epoch

Parameter-wise Plot

After epoch 100, plot:

- x-axis: Polynomial degree d
- y-axis: Final MSE

Q2. Comparison of L1 and L2 Regularization in Logistic Regression

Objective: To compare the behaviour of L1 and L2 regularization in binary classification.

Dataset Generation

Generate a dataset with at least 600 samples:

$$x_1, x_2, x_3, x_4, x_5 \sim U(-1, 1)$$

Define:

$$z = 2x_1 - 3x_2 + 0.5x_3^2 - x_4x_5 + \epsilon$$

$$\epsilon \sim \mathcal{N}(0, 0.2)$$

Assign labels:

$$y = \begin{cases} 1, & z > 0 \\ 0, & \text{otherwise} \end{cases}$$

Common Experimental Setting

- Train-test split: 70%–30%
- Optimization method: Gradient Descent
- Number of training epochs: 100 (fixed)
- Metrics must be recorded at every epoch

Part A: Without Regularization

- Train logistic regression without regularization.
- Plot:
 - Training accuracy vs epoch
 - Testing accuracy vs epoch

Part B: L2 Regularization (Ridge)

Train logistic regression models with:

$$\lambda \in \{0, 0.1, 0.2, \dots, 1.0\}$$

Plot:

- x-axis: λ
- y-axis: Final testing accuracy (after 100 epochs)

Part C: L1 Regularization (Lasso)

Train logistic regression models with the same λ values.

Plot:

- x-axis: λ
- y-axis: Final testing accuracy

Additionally:

- Record the number of non-zero coefficients for each λ .

Q3. Effect of Class Priors in Gaussian Classification

Objective: To study how changing class prior probabilities affects classification results and decision boundaries.

Dataset Generation

Using Python, generate a synthetic two-dimensional dataset:

- Generate 400 samples for Class 1 with mean $[1, 1]$.
- Generate 400 samples for Class 2 with mean $[4, 4]$.
- Use the same covariance matrix for both classes:

$$\begin{bmatrix} 1 & 0.4 \\ 0.4 & 1.5 \end{bmatrix}$$

Tasks

1. Equal Prior Case

- Train a Gaussian Naïve Bayes classifier assuming equal class priors.
- Plot:
 - Data points
 - Decision boundary
- Classify the test point $[2.5, 2.8]$.
- Compute:
 - Training accuracy
 - Testing accuracy
 - Confusion matrix

2. Unequal Prior Case

- Modify class priors to:
$$P(C_1) = 0.8, \quad P(C_2) = 0.2$$
- Re-train the classifier.
- Plot the new decision boundary.
- Classify the same test point $[2.5, 2.8]$.
- Compute:

- Training accuracy
- Testing accuracy
- Confusion matrix

3. Comparison and Analysis

- Compare the two decision boundaries.
- Explain how changing priors affects classification results.
- Discuss the impact on false positives and false negatives.
- Comment on real-world scenarios where class imbalance is important.

Submission Requirements

Submit:

- Source code
- All required plots
- Short analysis report